

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Applied Systems & Evaluation Aug 2025 to May 2026

- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.
- Boosted malware classification to **80.2%** across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.

Turito | Software Engineering Intern, Reliability & Customer Impact Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C++17, Rust

AI Systems: LLM routing and failover, RAG with source citations and abstention, agent orchestration, human-review routing, evaluation harnesses

Product & Integration: FastAPI, React, REST, gRPC, webhooks, Kafka, Temporal, Docker, AWS

Reliability: failure classification, retries and timeouts, structured telemetry, load testing, root-cause analysis

PROJECTS

AI Learning Hub, Grounded RAG | *Python, FastAPI, LangChain, OpenAI, Docker, AWS*

- Grounded answers over **10,000+** articles with source citations and abstention when retrieved evidence was insufficient.

Document Action Agent | *React, TypeScript, FastAPI, LLMs*

- Built a document-grounded assistant that verifies sources before recommending actions and routes ambiguous cases to **human review** instead of completing them silently.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

OPEN SOURCE

- microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)
- NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, Advanced Algorithms, Distributed Computing, Database Systems, GPU Programming

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)